

HORNSBY GIRLS HIGH SCHOOL

MATHEMATICS

YEAR 12 TRIAL EXAMINATION

2001

*Time Allowed – 3 hours
(Plus 5 minutes reading time)*

DIRECTIONS TO CANDIDATES

- a) Attempt all 10 questions.
- b) Start each question on a new page.
- c) All questions are of equal value.
- d) All necessary working should be shown. Marks may be deducted for careless or badly arranged work.
- e) Board-approved calculators may be used.

Question 1

- a) Evaluate $\left[\frac{\sqrt{3.12 + 6.9}}{5.03 - 2.9} \right]^3$ correct to two decimal places 1
- b) Evaluate $2|-2|^2 - 2^0$ 1
- c) Rationalise the denominator of $\frac{\sqrt{5} - \sqrt{2}}{\sqrt{5} + \sqrt{2}}$ 2
- d) Express $3x^2 + 27x + 60$ as the product of three factors 2
- e) Solve $6 - (x - 4) = x$ 2
- f) The value of a new car decreased by 12% or \$1500 in one year. What was the original value of the car? 2
- g) If $v^2 = u^2 + 2as$ find all possible values of u when $v = 35$, $a = 9.8$ and $s = 25$ (correct one decimal place). 2

Question 2 (Start a new page)

(a) Differentiate

5

i) $8x^5 - 7x^{-5}$

ii) $\sin 5x$

iii) $\frac{2x}{\log_e 2x}$

(b) Evaluate

5

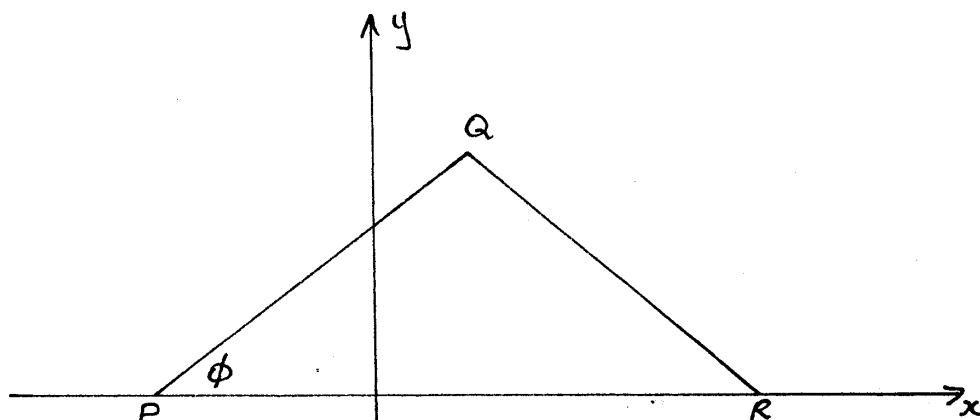
i) $\int_0^5 4e^{2x} dx$

ii) $\int_{-1}^1 (2x+7)^4 dx$

(c) Find $\int \frac{5y}{y^2+8} dy$

2

Question 3 (Start a new page)



In the diagram P , Q and R have coordinates $(-5,0)$, $(1,7)$ and $(7,0)$ and $\angle QPR = \phi$.

a) Find the equation of the line PQ in general form

2

b) Find the mid-point, M , of the interval QR

1

c) Find the distance PQ

2

d) Find the perpendicular distance from the point M to the line passing through the points P and Q

2

e) Find the exact area of the triangle PQM

2

f) Find the value of $\tan \phi$

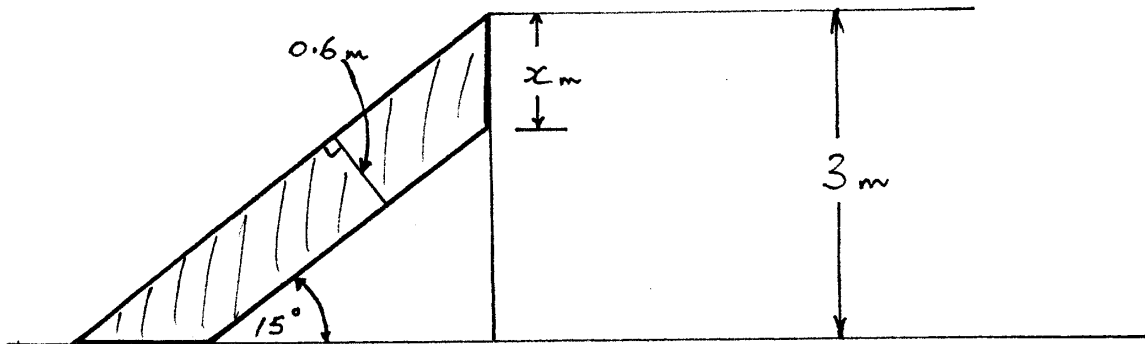
2

g) Prove that the point $(-2,4)$ does not lie on the interval PQ

1

Question 4 (Start a new page)

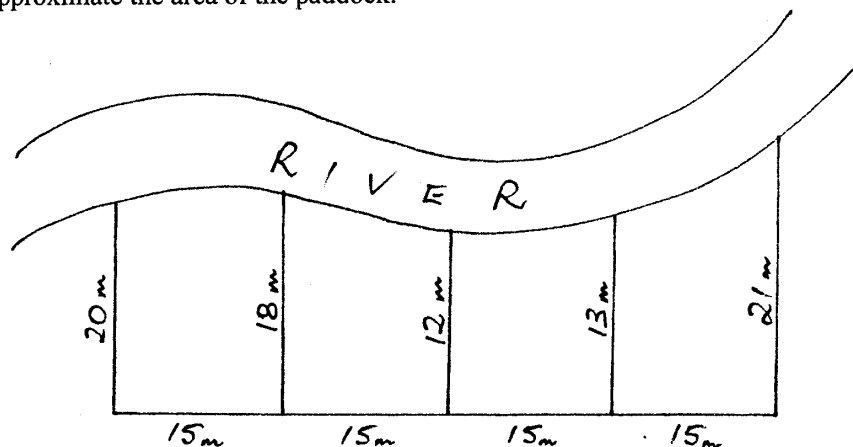
- A) The diagram shows a ramp, inclined at 15 degrees to the horizontal, that a builder is to build to allow cars to drive up a level, 3 metres high, in a car park. The cross-section of the ramp is in the shape of a trapezium and has been shaded on the diagram. The thickness of the ramp is 0.6 metres.



- a) calculate: 4
- The value of, x , on the diagram (correct to 3 decimal places)
 - Show that the area of the cross-section of the ramp is 6.2 m^2 (correct to 1 decimal place)
- b) The ramp is 5 metres wide and is to be made of concrete. What volume of concrete will be used to make this ramp ? 1
- B) The population of a small country town grows slowly at a rate proportional to its current population. The population exactly two years ago was 10 000 and is now 10 200.
- Find
- The growth constant of the population 2
 - The number of years, from now, that it will take the population to exceed 11 500 2
 - The population of the town in 5 years from today 1
- C) Solve $8^x = 4^{x-1}$ 2

Question 5 (Start a new page)

- a) The diagram below (not drawn to scale) shows a paddock with one side bounded by a river. Use Simpson's Rule with the five function values shown on the diagram to approximate the area of the paddock. 3



- b) When Jill was born her mother deposited \$180 into a Trust Account earning 12% p.a. interest compounded annually. She continued to deposit \$180 into this account each time Jill had a birthday. The last payment was made on Jill's sixteenth birthday. Calculate the total amount in the account on Jill's twenty-fifth birthday. 4
- c) Consider this arithmetic series $3 + 8 + 13 + 18 + \dots + 488$
- i) How many terms are in this series 1
 - ii) Find the sum of all the terms in this series 2
- d) For a particular series the sum to n terms is given by $S_n = 2^n + n^2$. What would be the tenth term of this series? 2

Question 6 (Start a new page)

- a) A particle starts from O and moves along a straight line so that after t seconds its distance from O is x cm, where

$$x = 6t - \frac{t^3}{2}.$$

- i) After how many seconds does it return to O and what is its velocity at that time? 2
 - ii) What is its distance from O when its velocity is zero and what is its acceleration here? 2
 - iii) What is its average velocity during the first two seconds of the motion? 2
- b) The continuous curve corresponding to the function $y = f(x)$ has the following properties in the closed interval $a \leq x \leq b$

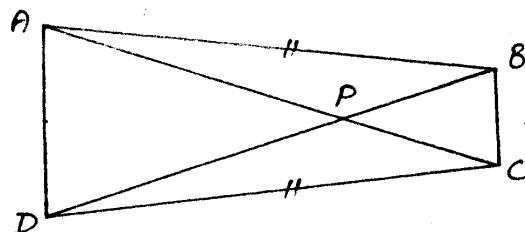
$$f(x) > 0, \quad f'(x) < 0, \quad f''(x) > 0$$

- i) Sketch neatly a curve satisfying these conditions. 2
 - ii) State the least value of $f(x)$ in this interval 1
- c) Find the stationary points and any points of inflexion of $y = x^3 - x^2 - x - 1$. 3

Question 7 (Start a new page)

- a) ABCD is a quadrilateral in which $AB = DC$ and $\angle BAC = \angle BDC$.
P is the point of intersection of the diagonals.

5



Prove that:

- i) $\triangle APB \equiv \triangle DPC$
 - ii) $\triangle PBC$ is isosceles
- b) A parabola has equation $y^2 - 6y + 25 = 8x$. Express this in the form $(y - p)^2 = 4a(x - q)$ and hence find:
- i) The coordinates of its vertex
 - ii) The equation of its axis of symmetry
 - iii) Its focal length
 - iv) The coordinates of its focus
 - v) The equation of its directrix

7

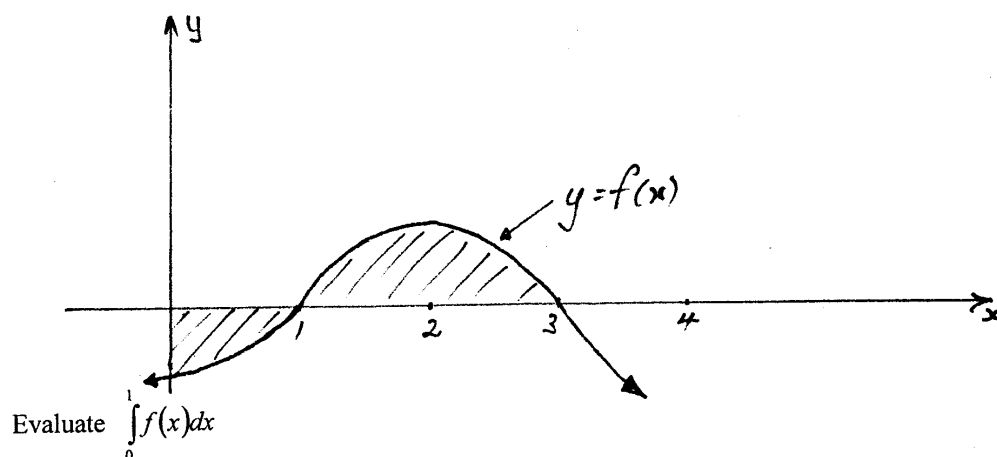
Question 8 (Start a new page)

- a) A athlete knows that she has a 20% chance of winning the 100 metre sprint event and a 30% chance of winning the 200 metres sprint. If she competes in both events at an athletics carnival what is the probability that she will:
- i) Win both events
 - ii) Not win both events
 - iii) Win one event only
 - iv) Win at least one event

4

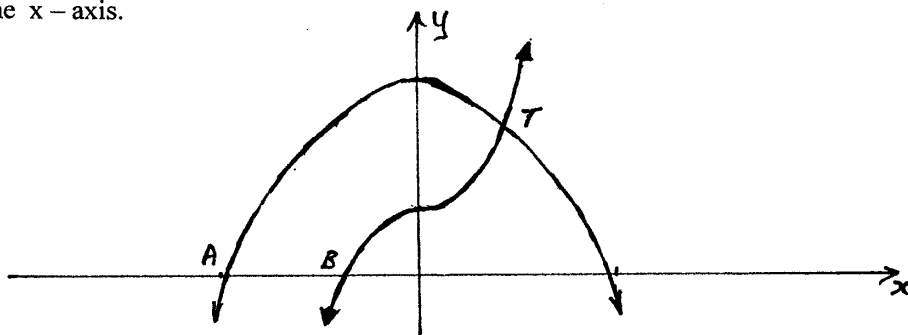
- b) In the diagram below $\int_1^3 f(x) dx = 5$ and the area of the shaded region is 7 units².

1



3

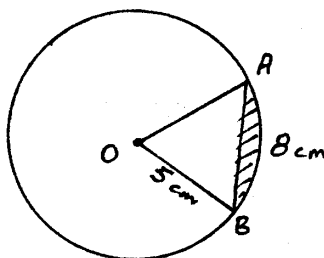
- c) The diagram below shows the shaded region enclosed by the curves $y = 3 - x^2$, $y = x^3 + 1$ and the x -axis.



- i) Find the coordinates of the points A and B. 2
- ii) Show that the point (T) of intersection between the curves is (1,2) 2
- iii) Hence find the area of the shaded region in exact form. 3

Question 9 (Start a new page)

- a) In the circle with center O, arc AB is 8cm and the radius OB is 5cm. Find: 3
 - i) The size of angle AOB, giving your answer in radians.
 - ii) The area of the shaded segment, giving your answer correct to 2 decimal places.



- b) The area under the curve $y = 1 - e^{-x}$, above the x -axis and between $x = 0$ and $x = 1$, is rotated about the x -axis. Prove that the volume of the generated solid is 3

$$\frac{\pi}{2} (4e^{-1} - e^{-2} - 1) \text{ cubic units}$$

- c) Sharon was driving a car in an off-road rally competition. From the start, S, Sharon drove 55 km due east to, A. At A, she proceeded on a bearing of 055° for 100 km to B. At B, she changed course to a bearing of 130° and continued in this direction until she reached the finish at C. (C is due east of A).

- i) Draw a diagram representing all this information on your answer sheet. 1
- ii) Show that angle ACB = 40° . 2
- iii) Find the distance from B to C. Give your answer to the nearest kilometre. 2
- iv) It took Sharon 24 minutes to travel from the start to the finish. What was her average speed in km/h? 1

Question 10 (Start a new page)

- a) A mechanic borrows \$P to buy new equipment for his business. The interest is compounded monthly at the rate of 18% p.a. The mechanic intends to repay the loan by making repayments of \$M per month (at the end of each month).

- i) Write an expression for the amount owed by the mechanic at the end of the first month ? 1
- ii) Write an expression for the amount owed at the end of N months. 2
- iii) If the mechanic had borrowed \$40 000 calculate the amount of the monthly repayment (\$M) if he wishes to repay the loan in 5 years. 3

- b) The mass, m grams, of a raindrop falling for, t, seconds in a humid cloud, is increasing at a rate

$$\frac{dm}{dt} \text{ where } \frac{dm}{dt} = \frac{1}{100} \left[t + \frac{t^2}{10} \right] \text{ gs}^{-1}$$

- i) If the initial mass of the raindrop is zero, what is the mass of the raindrop after 20 seconds ? 2
- ii) If the raindrop started as a smoke particle of mass 0.001 g, how much heavier would it be after 20 seconds than the raindrop in part (i) ? 1

- c) Observe that:

$$\begin{aligned} 1 &= 1 \\ 3x &= x + 2x \\ 5x^2 &= x^2 + 2x^2 + 2x^2 \\ 7x^3 &= x^3 + 2x^3 + 2x^3 + 2x^3 \\ 9x^4 &= x^4 + 2x^4 + 2x^4 + 2x^4 + 2x^4 \end{aligned}$$

By studying the above arrangement, or otherwise, find in simplest algebraic form, an expression for the limiting sum of the series

$$1 + 3x + 5x^2 + 7x^3 + 9x^4 + \dots + (2n-1)x^{n-1} + \dots \quad \text{3}$$

END OF PAPER

YEAR 12 TRIAL 2U 2001

- 1) a) $3 \cdot 28(21029)$
 b) 7
 c) $\frac{\sqrt{5}-\sqrt{2}}{\sqrt{5}+\sqrt{2}} \times \frac{\sqrt{5}-\sqrt{2}}{\sqrt{5}-\sqrt{2}}$
 $= \frac{5-2\sqrt{10}+2}{3}$
 $= \frac{7-2\sqrt{10}}{3}$
 d) $3x^2 + 27x + 60$
 $= 3(x^2 + 9x + 20)$
 $= 3(x+5)(x+4)$
 e) $6 - (x-4) = x$
 $6 - x + 4 = x$
 $10 = 2x$
 $x = 5$
 f) $12\% = \$1500$
 $1\% = \$125$
 $100\% = \$12500$
 g) $v^2 = u^2 + 2as$
 $3^2 = u^2 + 2 \times 9.8 \times 25$
 $u^2 = 735$
 $u = \pm 27.1 (10883)$
- 2) a) i) $40x^4 + 35x^{-6}$
 ii) $5 \cos 5x$
 iii) $\ln 2x \cdot 2 - 2x \cdot \frac{1}{x}$
 $= \frac{2 \ln 2x - 1}{(\ln 2x)^2}$
 $= \frac{2 \ln 2x - 1}{(\ln 2x)^2}$
 or $= \frac{\ln 4x^2 - 1}{(\ln 2x)^2}$
 b) i) $[2e^{2x}]_0^5$
 $= 2e^{10} - 2e^0$
 $= 2e^{10} - 2$ or $2(e^{10} - 1)$
 or 44050.932

- ii) $\left[\frac{1}{10} (2x+7)^5 \right]_{-1}^1 = \frac{1}{10} (9^5 - 5^5)$
 $= 5592.4$
 c) $\int \frac{54}{y^2+8} dy = \frac{3}{2} \int \frac{24}{y^2+8} dy$
 $= \frac{3}{2} \ln(y^2+8) + C$
- 3) a) $\frac{4-7}{x-1} = \frac{0-7}{-5-1}$
 $-6y + 42 = -7x + 7$
 $7x + 6y + 35 = 0$
 b) $x = \frac{1+7}{2} \quad y = \frac{7+0}{2}$
 $= 4 \quad = 3\frac{1}{2}$
 $\therefore (4, 3\frac{1}{2})$
 c) $d^2 = 6^2 + 7^2$
 $d^2 = 36 + 49$
 $d^2 = 85$
 $d = \sqrt{85}$
 $= 8.8 (317609)$
 d) $d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$
 $= \frac{|7 \times 4 + 6 \times 3\frac{1}{2} + 35|}{\sqrt{7^2 + 6^2}}$
 $= \frac{28 + 21 + 35}{\sqrt{85}}$
 $= \frac{84}{\sqrt{85}}$
 $= \frac{7\sqrt{85}}{13} = 4.7 (555635)$
 e) $A = \frac{1}{2} \times \sqrt{78} \times \frac{42}{\sqrt{78}}$
 $= 21$
 f) $\tan \theta = \frac{7}{6}$
 $7x + 6y + 35 = 0$
 $7 \times 2 + 6 \times 4 + 35 = 0$
 $-14 + 24 + 35 = 0$
 $= 45 \neq 0$
 $\therefore$ not on line.

- 4) a) i) $\cos 15^\circ = \frac{0.6}{x}$
 $x = \frac{0.6}{\cos 15^\circ}$
 $= 0.62 (11657)$
 ii) $\sin 15^\circ = \frac{3}{H}$
 $H = \frac{3}{\sin 15^\circ} = 11.5911$
 $\sin 15^\circ = \frac{2.38}{h}$
 $h = \frac{2.38}{\sin 15^\circ} = 9.1956$
 $\therefore A = \frac{1}{2} (11.5911 + 9.1956) \times 0.6$
 $= 10.39335 \times 0.6$
 $= 6.236$
 $= 6.2$
 b) $V = 6.2 \times 5$
 $= 31 \text{ m}^3 (31.0 \rightarrow 31.2)$
- 5) a) $P = P_0 e^{kt}$
 $10200 = 10000 e^{2k}$
 $e^{2k} = 1.02$
 $2k = \ln 1.02$
 $k = 0.0099$
 b) $11500 = 10200 e^{0.0099t}$
 $0.0099t = \ln \left(\frac{11500}{10200} \right)$
 $t = 12.1 \text{ years}$
 $\therefore 13 \text{ years}$
 c) $P = 10200 e^{5 \times 0.0099}$
 $= 10717.605$
 $\therefore 10717 \text{ or } 10718$
- 6) i) $x = 6t - \frac{t^3}{2}$
 $\frac{dx}{dt} = 6 - \frac{3t^2}{2}$
 when $x=0 \quad 6t - \frac{t^3}{2} = 0$
 $12t - t^3 = 0$
 $t(12 - t^2) = 0$
 $\therefore t=0 \text{ or } \sqrt{12}$
 when $x=\sqrt{12} \quad v = 6 - \frac{3 \times 12}{2}$
 $= -12$
 $\therefore$ returns when $t=\sqrt{12} \quad v=-12$.

$$ii) V = 6 - \frac{3t^2}{2}$$

$$\text{when } V=0, \frac{3t^2}{2} = 6$$

$$3t^2 = 12$$

$$t^2 = 4$$

$$t = 2$$

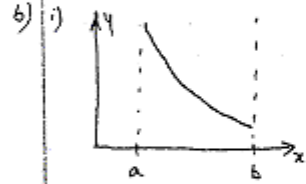
$$\text{when } t=2, x = 6 \times 2 - \frac{2^3}{2} = 8$$

$$a = -3t$$

$$\text{when } t=2, a = -6$$

$$iii) S = \frac{D}{T}$$

$$S = \frac{8}{2} = 4 \text{ cm/s}$$



$$ii) f(b)$$

$$c) y = x^3 - x^2 - x - 1$$

$$\frac{dy}{dx} = 3x^2 - 2x - 1$$

$$\frac{dy}{dx} = 6x - 2$$

$$\text{stat pt when } \frac{dy}{dx} = 0$$

$$\text{i.e. } 3x^2 - 2x - 1 = 0$$

$$(3x+1)(x-1) = 0$$

$$3x+1 = 0 \quad x = -\frac{1}{3}$$

$$x = -\frac{1}{3}$$

$$\text{when } x = 1, y = 1 - 1 - 1 - 1 = -2$$

$$x = -\frac{1}{3}, y = (-\frac{1}{3})^3 - (-\frac{1}{3})^2 - (-\frac{1}{3}) - 1 = -\frac{22}{27}$$

$$\text{pt inflexion when } \frac{d^2y}{dx^2} = 0$$

$$\text{i.e. } 6x - 2 = 0$$

$$6x = 2$$

$$x = \frac{1}{3}$$

$$\text{when } x = \frac{1}{3}, y = (\frac{1}{3})^3 - (\frac{1}{3})^2 - (\frac{1}{3}) - 1 = -1\frac{17}{27}$$

$$\therefore \text{stat pt } (1, -2) \text{ + } (-\frac{1}{3}, -\frac{22}{27})$$

$$\text{inflexion pt } (\frac{1}{3}, -1\frac{17}{27})$$

$$7) i) \text{ In } \Delta APB \text{ + } DPC$$

$$AB = DC \text{ (given)}$$

$$\angle BAC = \angle DAC \text{ (given)}$$

$$\angle APB = \angle DPC \text{ (vert opp)}$$

$$\therefore \Delta APB \cong \Delta DPC \text{ (AAS)}$$

$$ii) \text{ In } APBC$$

$$PB = PC \text{ corresponding sides of congruent } \Delta s.$$

$$b) y^2 = 6y + 25 = 8x$$

$$y^2 - 6y + 9 = 8x - 16$$

$$(y-3)^2 = 8(x-2)$$

$$i) (2, 3)$$

$$ii) y = 3$$

$$iii) 2$$

$$iv) (4, 3)$$

$$v) x = 0$$

$$8) i) .2 \times .3 = 0.06 \quad 6\%$$

$$ii) .8 \times .7 = 0.56 \quad 56\%$$

$$iii) .2 \times .7 + .8 \times .3 = .14 + .24 = 0.38 \quad 38\%$$

$$iv) 1 - 0.56 = 0.44 \quad 44\%$$

$$b) \therefore -2.$$

$$c) i) \text{ when } y = 0$$

$$3 - x^2 = 0$$

$$x = \pm\sqrt{3}$$

$$\therefore A = (-\sqrt{3}, 0)$$

$$x^3 + 1 = 0$$

$$x^3 = -1$$

$$x = -1$$

$$\therefore B = (-1, 0)$$

$$ii) 3 - x^2 = x^3 + 1$$

$$x^3 + x^2 - 2 = 0$$

$$\text{when } x = 1$$

$$1 + 1 - 2 = 0$$

$$0 = 0$$

$$\text{Thus } x = 1 \text{ is a soln.}$$

$$\text{when } x = 1, y = 3 - 1^2 = 2$$

$$\therefore (1, 2) \text{ is pt } T$$

$$iii) A = \int_{-\sqrt{3}}^1 (3 - x^2 - x^3 - 1) dx$$

$$= \int_{-\sqrt{3}}^1 (2 - x^2 - x^3) dx$$

$$= \int_{-\sqrt{3}}^1 (2x - \frac{x^3}{3} - \frac{x^4}{4}) dx$$

$$= (2x - \frac{x^3}{3} - \frac{x^4}{4}) \Big|_{-\sqrt{3}}^1$$

$$= (2 - \frac{1}{3} - \frac{1}{4}) - (-2\sqrt{3} + \frac{3\sqrt{3}}{2} - \frac{9}{4})$$

$$= 2 - \frac{1}{3} - \frac{1}{4} + \sqrt{3} + \frac{9}{4}$$

$$= \frac{11}{3} + \sqrt{3} \text{ or } 3\frac{2}{3} + \sqrt{3}$$

$$= (5.3987175)$$

$$9) a) i) L = 10$$

$$S = 50$$

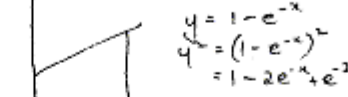
$$\theta = \frac{\pi}{3}$$

$$ii) A = \frac{1}{2} r^2 \theta - \frac{1}{2} r^2 \sin \theta$$

$$= \frac{1}{2} \times 25 \times \frac{\pi}{3} - \frac{1}{2} \times 25 \times \sin \frac{\pi}{3}$$

$$= 7.51 \quad (7.50533)$$

$$b)$$



$$V = \pi \int_0^1 y^2 dx$$

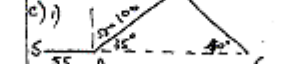
$$= \pi \int_0^1 (1 - 2e^{-x} + e^{-2x}) dx$$

$$= \pi [x + 2e^{-x} - \frac{1}{2}e^{-2x}]_0^1$$

$$= \pi [1 + 2e^{-1} - \frac{1}{2}e^{-2} - 0 - 2 + \frac{1}{2}]$$

$$= \pi (-\frac{1}{2} + \frac{1}{2}e^{-2} + 2e^{-1})$$

$$= \frac{\pi}{2} (4e^{-1} - e^{-2} - 1)$$



$$ii) \angle CBX = 130 - 90$$

$$= 40$$

$$\angle ACB = \angle CBX = 40 \text{ (alt } \angle s \text{ B} \parallel \text{AC)}$$

$$iii) \frac{BC}{\sin 35} = \frac{100}{\sin 40}$$

$$BC = \frac{100 \sin 35}{\sin 40}$$

$$= 89.2 \quad (326.53)$$

$$iv) S = \frac{89 + 100 + 55}{2}$$

$$= 610 \text{ (km/h)}$$

$$10) i) A_1 = 1.015P - M$$

$$ii) A_2 = 1.015^2P - 1.015M - M$$

$$A_3 = 1.015^3P - 1.015^2M - 1.015M - M$$

$$A_n = 1.015^n P - M(1.015^{n-1} + 1.015^{n-2} + \dots + 1.015^1 + 1.015^0)$$

$$iii) 1.015^{60} \times 40000 - M \left(\frac{1(1.015^{60} - 1)}{1.015 - 1} \right) = 0$$

$$M = \frac{1.015^{60} \times 40000 \times 0.015}{1.015^{60} - 1}$$

$$= \$1015.74 \quad (\$1015 - \$1016)$$

$$6) i) \frac{dm}{dt} = \frac{1}{100} \left[t + \frac{t^2}{10} \right]$$

$$m = \frac{1}{100} \left[\frac{t^2}{2} + \frac{t^3}{30} \right] + C$$

$$\text{when } t=0, m=0 \therefore C=0$$

$$\therefore \text{when } t=20$$

$$m = \frac{1}{100} \left[\frac{400}{2} + \frac{8000}{30} \right]$$

$$= 4\frac{2}{3} \text{ g or } 4.7 \text{ g or } 4.6 \text{ g}$$

$$ii) \text{when } t=0, m=0.001 \therefore C=0.001$$

$$\therefore m = \frac{1}{100} \left[\frac{t^2}{2} + \frac{t^3}{30} \right] + 0.001$$

$$\therefore 0.001 \text{ g}$$

$$c) S = \frac{1}{1-x} + \frac{2x}{1-x} + \frac{2x^2}{1-x} + \dots$$

$$S = \frac{1}{1-x} (1 + 2x + 2x^2 + \dots)$$

$$= \frac{1}{1-x} \left(1 + \frac{2x}{1-x} \right)$$

$$= \frac{1}{1-x} + \frac{2x}{(1-x)^2}$$

$$= \frac{1-x+2x}{(1-x)^2} = \frac{1+x}{(1-x)^2}$$